Breaking Down Context Rot in LLMs

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Abstract

Long conversations are said to wear language models down, but the standard evaluations confound *accumulation* (more context) with *dilution* (the answer-bearing tokens becoming a smaller, more distant, more contested fraction of it). We build a harness that separates them: streams of individually localized tasks in which context length, evidence distance, context confusability, and demonstrated answer shape are four independent variables, instrumented with a span-attention clamp that can *set* any span's post-softmax attention share. Four findings. (1) Accumulation alone shows no net cost: accuracy and instruction adherence are flat to 90% fill, with bounded nulls. (2) Displacing the evidence costs accuracy, and attention mass is *causally sufficient* for the cost: removing the evidence's share with nothing moved reproduces the full penalty, and the relationship between mass and accuracy is smooth, with no threshold. (3) Near-duplicate context costs further accuracy while the evidence's attention barely moves. Closing every context mention of the probe's answer options at generation time recovers 56% of that penalty, against a size-matched random-closure control at zero, so competition is attention-mediated too—through the *competitors*, not the evidence. (4) Accumulated filler that demonstrates an *applicable* answer shape erodes an instructed output format completely ($0.875 \rightarrow 0.000$ within three filler turns, accuracy unharmed), while the instruction's attention holds and its content stays linearly decodable. Attention mass is still a causal *weight* in that contest: clamping the instruction's share down collapses compliance where nothing competes, clamping it up restores compliance against 42 counter-exemplars, and restating the instruction restores it without surgery. The same closure that rescues competition restores nothing here: the demonstrated mode is already in place at prefill—and counterfactual patching localizes its carrier to the chat template's delimiter tokens, in the lower half of the stack. In these localized streams, context rot is not a decay of competence with length. It is displacement, competition, and precedent, and attention dilution proper is only the first.

1 Introduction

A recurring worry about long conversations is *context rot*: as turns accumulate, the model is said to degrade. The phenomenology is real. On a repeating multiple-choice task, instruction-tuned models grow more confident as context fills: next-token entropy drops roughly 3–4$\times$ on Qwen-2.5-7B (Yang et al., 2024) and Llama-3.1-8B (Grattafiori et al., 2024). On OLMo-2-Instruct the last token's attention leaves the system prompt (correlation with fill -0.93) and the current query (-0.71) for the most recent prior cases ($+0.89$), all monotone. By every visual diagnostic this is a strong effect that grows with context.

The tempting reading is that length wears the model down. But almost every context-rot evaluation embeds the answer-bearing information inside *one long task*—a long document, a buried needle, a growing codebase. Longer context then mechanically *dilutes* the relevant tokens among irrelevant ones, so a measured degradation conflates the cost of accumulation with the cost of the target being

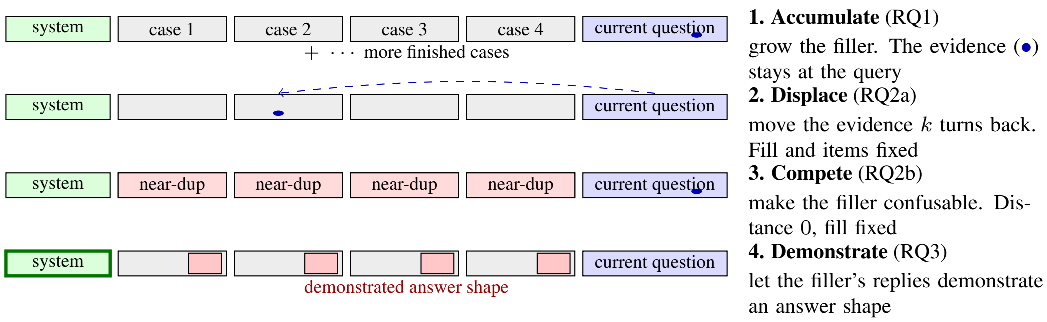

Figure 1: **The localized-stream framework: four independent knobs on one transcript.** Each stream accumulates self-contained cases, so the evidence for the current answer sits at the current question by default and every knob is turned with the others held fixed. The span-attention clamp (§2.3) additionally sets any span's attention share directly, turning the mediating variable itself.

harder to attend to ("lost in the middle," Liu et al., 2023). The two cannot be told apart in that setup, and the distinction matters in practice: an agent transcript of finished, locally-scoped steps accumulates without diluting, and the two accounts predict opposite outcomes for it (Appendix B).

We separate the candidate causes inside one harness (Fig. 1) and ask three questions. **RQ1**: does accumulation alone, with the evidence for every answer held at the current query, degrade accuracy or instruction adherence? **RQ2**: when dilution does cost accuracy, is the cost mediated by the evidence's attention mass? We split dilution into its two components, the evidence becoming *distant* and the context becoming *confusable*, and test each causally. **RQ3**: if adherence survives context fill, what erodes instruction-following under accumulation?

RQ3 engages Dongre et al. (2026) directly: they showed that attention from responses to the system prompt closes over accumulated filler and asked whether behavior survives that closure, finding the answer architecture-indexed. Our precedent arms take up their question with a graded instrument on one architecture.

Accumulation is the design's null arm (RQ1). The mechanisms that cost something are displacement, competition, and precedent. Our contributions are the harness that separates them and one causal result for each:

- A harness that makes accumulation and dilution independent, by accumulating individually localized tasks whose evidence never moves, and a span-attention clamp that *sets* any span's post-softmax share with a bit-identical no-op at scale 1 (§2).
- **Displacement** is causally sufficient: removing the evidence's attention mass with nothing moved reproduces the penalty in full, and the mass→accuracy relationship is smooth and graded with no threshold in it (RQ2, §4.2).
- **Competition** is a *second* mechanism, costing accuracy at constant *evidence* mass. Closing the competitor mentions recovers 56% of the penalty, so both accuracy mechanisms are attentional but on different spans, and span-targeted closure tells them apart (RQ2, §4.3).
- **Precedent** costs compliance rather than accuracy. Accumulated demonstrations erode an instructed format only when their shape transfers to the probe, immediately and with accuracy unharmed. The erosion is not carried by the instruction's attention, yet re-weighting or re-presenting the instruction each fully reverse it (RQ3, §4.4–§4.5).

2 Methodology

2.1 The localized-stream framework

Our harness accumulates **individually localized tasks**: a stream of self-contained 5-option medical cases (DDXPlus, Tchango et al., 2022) or discrete factual questions (MMLU, Hendrycks et al., 2021). Each answer depends only on the *current* question, which always sits at the end of the context. Prior turns are never required to answer it. Accumulation therefore grows the context without diluting the information the current answer needs, and each component of dilution becomes an *independent variable* (Fig. 1). Holding the item, the model, the total fill, and the metric fixed, we can move the evidence k user turns back (*displacement*), replace the filler with near-duplicate cases (*competition*), or let the filler's replies demonstrate a competing answer shape (*precedent*). Question text is byte-identical across arms, and the filler is identical unless it is the treatment. Distance, fill, confusability, and demonstration are inseparable in a long-document setup. Here they vary independently.

2.2 Span share and enrichment

For a token span A (the evidence, the system prompt, a filler answer), its *share* is the final position's post-softmax attention mass on A, averaged over heads and pooled over all 32 layers. The ladder and sweep diagnostics (Table 3, Fig. 2) additionally read layer 24 as a reference. A fixed-size span's raw share falls mechanically as context grows (a 48-token system prompt is a shrinking fraction of the key space), so a raw share decline says nothing by itself. We therefore also report *enrichment*: share divided by the span's token fraction, which is 1 for a span attended in proportion to its size. The distinction matters in §4.4: raw system share falls roughly an order of magnitude on the same curve in arms that erode and arms that do not, so it separates nothing. Metrics that aggregate raw attention-to-goal share across a growing context, such as the goal-accessibility ratio of Dongre et al. (2026), inherit this arithmetic decline by construction.

2.3 The span-attention clamp

To intervene on attention rather than position, we use a clamp that *sets* a span's post-softmax share: a constant bias b added to the span's key columns in the additive attention mask scales the span's odds by exactly e^b, and softmax renormalizes. Nothing in the attention forward is reimplemented, so a scale of 1.0 (b=0) is bit-identical to the unhooked model. A bisection on b hits any target share at the reference layer to $\sim 10^{-4}$, and scale 0 removes the span from attention entirely. We call this *closing* the span, and use the term throughout. Closure is the hard-masking intervention of Dongre et al. (2026), recovered as the clamp's endpoint, so their ablation and our graded doses sit on one dial. The clamp is graded and bidirectional: it can remove a span's mass or restore it, so mediation can be tested in both directions rather than only ablated. It is uniform across heads and, unless stated, applied at every layer. Per-head structure is measured separately (Appendix G). Unlike attention-steering methods aimed at improving compliance (Zhang et al., 2024), the clamp is a measurement instrument: its no-op is exact, its dose is solvable, and its near-ablation regime is excluded from interpretation (§4.2). Validation is in Appendix C.

2.4 Residual probes and mode-vector steering

To ask what the residual stream represents while behavior changes, we train linear probes (Alain and Bengio, 2016; Belinkov, 2022) on the final pre-generation position across all 33 stack rows, with episode-level leave-one-out cross-validation and permutation nulls. To test what a decodable direction *does*, we steer with mean-difference vectors at decode time against norm-matched random controls, and erase by projection. Full protocol in Appendix I.

3 Experimental design

Streams and probes. All causal experiments run on OLMo-2-7B-Instruct at seed 42, greedy decoding. Appendix K replicates the program on Qwen2.5-7B-Instruct. Every mechanism experiment reported here runs to a 4k window, on both families, so that no arm comparison is confounded by window size. The exceptions are not mechanism results. The adherence arm in §4.1 runs to 32k, and the WildChat measurements behind the filler design run to 16k and 32k. A 4k window holds only tens of cases, so we pool many independent accumulation sessions. A reversed-question-order control removes difficulty-ordering confounds. Probe options are shuffled per item and items with fewer than 5 options are excluded (DDXPlus lists differentials in rank order, so unshuffled gold letters are 71.4% "A"—a letter-bias trap for any arm comparison, Appendix D).

Compliance instrumentation. For RQ3 the system prompt carries a clinical answer template (`ANSWER:` letter, then `SUPPORTING:` with at least two vignette findings) and every reply is graded separately for compliance and accuracy by a rule-based checker. Simpler canary instructions (a reply

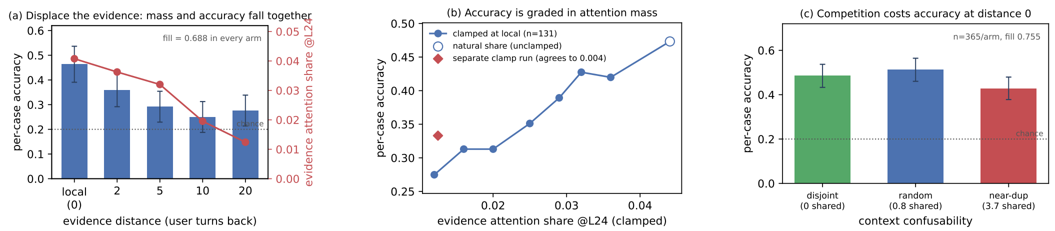

Figure 2: **Dilution costs accuracy. Accumulation does not.** (a) Moving the evidence k user turns back, at identical fill and byte-identical questions, drains its attention mass and costs accuracy together. (b) Setting that mass directly, with the evidence left in place, reproduces the penalty and traces a smooth graded dose-response—no threshold. The diamond is an independently-run clamp experiment measuring the same endpoint cut. The two agree to 0.004 on this panel's layer-24 indexing (0.017 on Table 1's all-layer scale). (c) At distance 0 and fixed fill, near-duplicate accumulated context costs a further 0.085. Bootstrap CIs resample cases throughout.

prefix and suffix) instrument adherence in the accumulation and clamp arms. Grader rules, filler construction, and the three filler streams' demonstrated shapes are specified in Appendix D.

Statistics. Accuracy is reported as a fraction throughout. "Points" are percentage points (1 point $= 0.01$ of accuracy), the scale of the joint-fit coefficients (Appendix F). Per-condition n is bounded by the context window, so every clamp experiment is paired over shared items and analysed with a paired bootstrap resampling *cases* (10,000 draws), never heads or turns. Resampling arms independently inflates intervals about $2.5\times$. An overflow guard skips rather than truncates every item that would not fit whole, because truncating a long item near the window edge manufactures exactly the late-window cost under study. It reports per-arm skip counts (Appendix F).

4 Results

4.1 Accumulation shows no net cost

Per-case accuracy does not fall as context fills. It is worst at the cold start and warms up ($\mathrm{corr}(\mathrm{correct}, \mathrm{fill}) = +0.11$ Instruct, $+0.07$ DPO). A homogeneous repeating task could let in-context learning mask a hidden decay, so we test that directly: in a *random-subject* stream each question is drawn uniformly from all 57 MMLU subjects, so prior context carries no predictive information about the next question—only the answer *format* is learnable. Accuracy is still flat: its correlation with fill is -0.02 (random, n=728) and $+0.032$ (coherent, n=988), computed over fill < 0.8, and the nulls are *bounded*—at 95%, hidden declines larger than 2.9 points (coherent) or 5.5 (random) are excluded. Checkable instruction adherence (the canary instructions) stays at ceiling throughout, in the program's one adherence arm, which ran on Qwen2.5-7B-Instruct to a full 32k window. That null is narrower than it looks: the canaries were easy and the accumulated history is itself compliant, so it shows adherence surviving *compliant* history. §4.4 shows it does not survive history that demonstrates a competing format. Accumulation also drives the current query's attention share down to ≈ 0.15, the level at which clamping the same share on a cold-start context begins to cost accuracy (Appendix H). That accuracy stays flat anyway is consistent with in-context learning compensating for the lost attention. The accuracy null replicates on Qwen2.5-7B (Appendix K).

4.2 Displacement acts through attention mass

We now vary only *where the answer-bearing evidence sits*. Each DDXPlus probe's vignette is placed k user turns before its question. Mean fill is 0.688 in every arm, the question text is byte-identical, and the overflow guard skipped 0 of 192 probes (an example transcript with its generations is in Appendix E). Every gap's CI excludes zero (Table 3, Fig. 2a). In a joint fit of accuracy on fill and distance, distance carries the coefficient ($\beta = -0.0076$ $[-0.0117, -0.0035]$, significant) and fill carries none ($\beta = -0.007$ $[-0.210, +0.187]$). Within the `local` arm, the one that reproduces the accumulation design, accuracy is flat with fill ($\beta = -0.294$ $[-0.767, +0.184]$). The effect survives restriction to parsed responses and is then fully monotone ($0.524 \to 0.413 \to 0.397 \to 0.343 \to 0.338$). One qualification: the decline saturates by $k{\approx}10$ rather than deepening to k=20, and those two arms are not distinguishable from each other. So the same model, on the same items at the same

Table 1: **The mass interventions** (all-layer shares, paired bootstrap 95% CIs. Penalty fractions are against the same panel's own displacement penalty.

intervention	contrast	Δ accuracy	of the penalty
removal (`local` $\to$ `back_20` share)	`local` $-$ clamped	$+0.151$ $[+0.099, +0.208]$	0.91 $[0.60, 1.46]$
	clamped $-$ `back_20`	$+0.016$ $[-0.052, +0.083]$	lands on `back_20`
sweep endpoint (displaced share)	natural $-$ endpoint	$+0.168$ $[+0.102, +0.234]$	matches removal
restoration (`back_20` $\to$ `local` share)	restored $-$ `back_20`	$+0.047$ $[+0.010, +0.083]$	0.28 $[0.07, 0.61]$
restoration, per-head pattern	per-head $-$ uniform	-0.021 $[-0.083, +0.042]$	no gain

fill, degrades when the evidence is displaced and does not degrade when it is not. The localized null is not a ceiling artifact: these items *can* show degradation, and do, as soon as dilution is reintroduced.

Distance and attention drain move together by construction, so Table 3 is consistent with dilution but does not establish it—a purely positional account predicts the same table. Four interventions separate them.

Observational correlations. Evidence share falls $0.0408 \to 0.0124$ across the ladder ($r = -0.83$ with distance) while the *question*'s share stays flat, confirming only the evidence moved. One caution: *within* an arm, higher evidence share predicts *lower* accuracy ($\beta = -11.2$ $[-20.0, -2.6]$, stronger controlling for vignette length). Attention there tracks item difficulty (the model looks harder at confusing vignettes) and has the *opposite sign* to the causal effect below. Reading share–accuracy correlations off this data without intervening yields the wrong sign.

Mass removal. Keeping the evidence at `local` and clamping its share down to the *same item's own* `back_20` share reproduces the displacement penalty and lands the arm on `back_20` (Table 1, paired n=192). **Mass removal alone reproduces the displacement penalty, with nothing moved.**

Dose-response. Sweeping the evidence's share at fixed position over seven levels (layer-24-indexed, common subset n=131 present at every level) gives a graded decline from 0.473 at the natural share (0.0441) to 0.275 at 0.012 (Fig. 2b). Six of the seven levels differ significantly from the natural share, including a cut of less than a fifth of the mass, and no step is a threshold: the largest adjacent step is 0.053 and the decline is smooth throughout. Its endpoint at the displaced share matches the independently-run removal experiment in both denominations—to 0.004 layer-24-indexed, 0.017 on Table 1's all-layer scale. The relationship is *graded*: a slope, not a cliff.

Mass restoration. The converse is only partial: clamping the displaced evidence's share back *up* to its `local` level recovers little of the penalty (Table 1), leaving most of it in place. The natural suspect is the instrument: a uniform clamp strips mass faithfully but cannot reconstruct *which* heads should carry the restored mass. A pattern-matched test rules that out: one closed-form bias per query head per layer, restoring each item to its own donor's per-layer, per-head pattern (installed to a mean per-head share error of 0.004, bias SD 1.22 across heads) gains nothing over the uniform clamp (Table 1). Head identity is not what restoration is missing. What no per-head clamp restores is the distribution *within* the span. One bias per head is still uniform over the evidence's $\sim$280 tokens, so the unrecovered remainder is either that token-level pattern or a positional channel outside attention altogether. The drain itself is positional in *tokens*, not turns. A 64% mass cut from token distance alone costs at most 9.4 points, consistent with RoPE distance decay (Su et al., 2024; Wu et al., 2025) rather than conversation structure (Appendix H). The ladder, the sufficiency clamp (107% of the gap), and the graded dose-response all replicate on Qwen2.5-7B, there with no unparsed replies (Appendix K).

4.3 Competition acts through the competitors, not the evidence

Burying a needle varies two things at once: the evidence becomes distant *and* it becomes surrounded by plausible alternatives. We isolate the second by holding the evidence at the current query and fill fixed, and varying only how *confusable* the accumulated context is. Every arm accumulates eight prior DDXPlus cases, each answered in the transcript with its gold letter, so format and in-context-learning affordance are constant. The arms differ only in how much the accumulated cases' differentials overlap the probe's five options: `random` (the natural stream, 0.80 shared options on average), `disjoint` (zero), and `near_dup` (at least 3, averaging 3.75). No arm admits a context

Table 2: **Generation-time closures on the near-duplicate arm** (paired n=365). Every arm closes the same $\approx$128-token budget. Net = closure − its size-matched random control.

closed set	mass closed	net accuracy [95% CI]
verbatim option-name mentions	0.0077	$+0.060$ $[+0.008, +0.112]$
verbatim, re-run alongside the arms below	0.0076	$+0.069$ $[+0.016, +0.121]$
most-read content tokens	0.087	$+0.003$ $[-0.049, +0.055]$
most-read tokens including template glue	0.416	-0.175 $[-0.230, -0.121]$

case with the probe's own gold, so the manipulation is that in `near_dup` the probe's gold routinely appears as some context case's *distractor* (Appendix D). Paired over n=365 probes, accuracy is `random` 0.512, `disjoint` 0.485, `near_dup` $\mathbf{0.427}$ (Fig. 2c).

Near-duplicate context costs 0.085 $[0.030, 0.140]$ against the natural stream and 0.058 $[0.006, 0.112]$ against disjoint context. In a joint fit within this experiment, option overlap carries the coefficient ($\beta = -0.021$ $[-0.039, -0.003]$) and fill carries none (β n.s.)—the same shape as the distance result, on a different variable. Two controls hold: the low-competition arms land on §4.2's `local` accuracy of 0.464 (both gaps n.s.), and context length does not order the arms— `disjoint` is the *longest* and mid-accuracy while `near_dup` is the shortest, so length biases against the result. The effect survives restriction to parsed responses ($+0.082$, significant), and it is *not* monotone in confusability: `random` sits above `disjoint`, not below it (n.s.). We therefore claim that near-duplicate context is costly, *not* that cost rises with overlap—one significant step, not a gradient.

The dissociation on mass. Repeating the sweep with attention capture (same probes, same seeds) separates the two mechanisms on mass: displacement removes 0.0455 of the evidence's attention share and competition removes 0.0022, while costing 0.188 and 0.085 accuracy respectively—a $\sim$20$\times$ smaller mass change at half the accuracy cost, far too small to act through the displacement dose-response. The two drain profiles are *negatively* correlated across layers ($r = -0.32$): displacement bites hardest at layers 0–3 and 16, competition at 17–18—the same accuracy contrast with opposite attention signatures, evidence for two mechanisms that no single-layer measurement could supply. Per-head structure, including the null result that no head hides a large countervailing move, is in Appendix G.

The competitor-closure test. An evidence-mass change this small rules out starvation as the carrier. It does not touch the other attentional route, in which the filler's (arm-constant) mass lands on *instances of the probe's own answer candidates* and reading them is what misleads. We test this directly: closing every context occurrence of the probe's option names at generation time recovers **56%** of the competition penalty (Table 2), the residual gap to `random` no longer excludes zero, and the rescue survives restriction to parsed responses. **So competition is attention-mediated after all—through the competitors, not the evidence**: less than 1% of the attention distribution carries the effect, and removing it removes most of the cost. Both accuracy mechanisms are therefore attentional, on different spans: displacement removes mass from the evidence, competition directs mass onto the competitors, and the evidence-mass measurement dissociates them because it is blind to the second by construction. The $\sim$40% residual is not instrument slack of the high-attention kind: closing the most-read *content* tokens of the context body (11$\times$ the mentions' mass) recovers nothing (Table 2). The closure works because of what the closed tokens *say*, not the mass they receive, and the residual accordingly points at prefill-borne interference. The same measurement concentrates 0.42 of the *entire* final-position row in the $\approx$128 most-read tokens, dominated by the chat template's turn delimiters. Closing *that* set is broadly destructive, but what it disrupts is the demonstrated reply shape, not competition (§4.5).

The family divergence. Competition is the one mechanism that does not carry across families, and the divergence is itself informative. On the same n=365 panel Qwen2.5-7B shows no detected penalty (`random` − `near_dup` = $+0.030$ $[-0.016, +0.074]$), and the scale-0 competitor closure that recovers 56% of OLMo's penalty does nothing there ($+0.019$ $[-0.025, +0.063]$, random-closure control 0.000). The attention signatures separate with disjoint intervals. Under near-duplicate context OLMo's evidence share is flat ($+0.0003$, null) while Qwen's *rises* ($+0.0071$ $[+0.0062, +0.0080]$), as if that model defends the evidence under confusability rather than merely not paying for it (Fig. 8c, d). The two families therefore do not differ by degree on one mechanism. They differ in whether the competitor-reading channel is engaged at all, so we treat competition as architecture-indexed and rest its positive claims on OLMo (Appendix K).

4.4 Precedent: in-context learning against the system prompt

The costs so far are to accuracy. A third mechanism costs *compliance*: not whether the model can still solve the task, but whether it answers in the instructed form (§3, n=40 probes per depth, one accumulation snapshotted at every depth, so depth is the only mover).

Every stream here is also a stream of in-context examples. Each prior case is answered in the transcript, so the history *demonstrates* an answer shape while the system prompt *instructs* one. Two channels, one decision. When they agree, as in §4.1's compliant history, nothing surfaces. This section makes them disagree, and the instruction is what gives way (cf. Anil et al., 2024).

The instruction's attention is causally necessary—and demonstration masks it. Accumulation drives the system span's share down $8\times$ within eight turns ($0.166 \rightarrow 0.021$). In a context containing no assistant turn, clamping that share from 0.165 to 0.050 collapses compliance from 0.99 to 0.03 (a suffix canary flips on 120 of 120 items) while accuracy is unchanged ($0.525 \rightarrow 0.467$, n.s.) and parsing holds at 0.967: compliance is destroyed and competence is not. The clamp level is one accumulation passes on its own between two and four turns (-2.6 nats), well clear of the near-ablation regime excluded in §4.2. This confirms Dongre et al. (2026)'s ablation finding that system-prompt attention is causally necessary, and sharpens it: the collapse needs no masking, only a graded clamp to a share accumulation reaches by itself. But one *compliant* example in context pins compliance at ceiling at every clamp level, and one bare-letter example pins it at floor even unclamped—replies byte-copy the previous assistant turn's format, with zero length variance across 720 generations per arm (Appendix I). The instruction governs behaviour only when the transcript is silent. This re-reads §4.1's adherence null: that null accumulates *compliant* history, so it shows compliance survives compliant precedent, not accumulation.

Erosion tracks applicability, not attention. If the contest is with in-context learning, what should decide it is how far the demonstrated shape *transfers* to the probe. Three filler streams vary exactly that: `code` (request→Python function, no shape applicable to a 5-option probe), `gsm8k` (worked prose, loosely applicable), `mmlu` (MCQ→bare letter, a drop-in replacement). `code` never erodes (compliance 0.875 at cold start, 1.00 at matched fill 0.5 and 0.900 at fill 0.778, the deepest interpreted point). `gsm8k` erodes late and partially (0.600 at fill 0.48, replies drifting into the filler's worked-solution style). `mmlu` collapses completely and immediately: $0.875 \rightarrow 0.000$ at three filler turns (fill 0.147), never recovering through fill 0.87 (Fig. 4a). At matched fill $\approx$0.5 the arms order $1.00/0.60/0.00$. Accuracy on the same replies is unharmed everywhere (0.425 shallow, 0.500–0.684 deep): the model still solves the case—it answers in the demonstrated style. The system prompt's attention does not carry the difference: its raw share falls on the same arithmetic curve in every arm, its per-token enrichment is flat-to-rising in all three (Fig. 4b), and `code` holds full compliance at half the raw share at which `mmlu` has already collapsed. This answers the survival question of Dongre et al. (2026) on an axis their fixed-filler design cannot see: the same model at the same near-zero share survives or collapses depending on what the filler demonstrates.

Reversal is total, by either lever. At depth 42 (natural compliance 0.000, accuracy 0.675), clamping the system span back to its cold-start share restores compliance to 1.000 at an accuracy cost of $0.675 \rightarrow 0.425$: mass forced onto a 48-token span is taken from everything else, §4.2's zero-sum arithmetic. Restating the instruction in the latest user turn also restores 1.000, at accuracy 0.450 and with no intervention on the model. Both levers together give compliance 1.000 at accuracy 0.275 (Fig. 4c). Natural dilution therefore does not cause the erosion—it is identical in the arm that never erodes. An applicable competing template does. The outcome is a contest in which attention mass is a causal *weight*: clamp the instruction down and it loses unopposed, clamp it up and it wins against 42 counter-exemplars, leave it natural and the demonstrated template beats it. Re-presenting beats re-weighting, though both work. The applicability ordering, the reading signature, and restatement's advantage all replicate on Qwen2.5-7B, where the erosion is staged rather than total (Appendix K).

4.5 The mode: readable, installable, not erasable

We call the behavioral state §4.4 leaves behind the *mode*: the disposition to answer in the demonstrated shape rather than the instructed one. Where is it, if not in attention? The demonstrated answer spans *are* preferentially read exactly where erosion occurs (per-token enrichment of filler answers over filler questions: $+1.28$ in `mmlu` and $+0.34$ in `gsm8k` at peak, negative in `code`—the erosion ordering). But reading them is not what sustains the erosion: masking every exemplar answer's key columns at generation time restores nothing (0.000 compliant, and a size-matched random-masking control is equally null). Nor is it what *installs* it: splitting the closure by window (active during prefill only, released at decode, or the converse) leaves compliance at 0.000 in both windows, each against a size-matched random control at its own window (n=40, depth 42). The split instead localizes what exemplar-answer reading is worth to *accuracy*: its entire contribution is prefill-borne—closing during prefill costs 0.175 $[0.075, 0.300]$ of accuracy, closing during decode costs exactly nothing item-for-item, and closing both windows is item-for-item identical to closing prefill alone. The privileged reading is consistent with a template already in place by prefill, task-vector style (Hendel et al., 2023)—installed through some channel other than attention to the answer spans. The contrast with §4.3 is the sharpest dissociation in the paper: the *same* scale-0 closure that recovers 56% of the competition penalty restores nothing here, in any window (Fig. 5). Competition is generation-time *reading*. The precedent mode is in place *by prefill*.

Residual-stream probes (§2.4) complete a three-level dissociation. The instruction's presence remains linearly decodable at every depth (transfer AUC 1.000, permutation $p<0.005$) while it has zero behavioral force: read, represented, overruled. Whether the upcoming reply will comply is decodable *before the first token* (leave-one-out AUC 0.875 at layer 22) within cells matched on depth, fill, and filler. The mode is installable by direction: adding the mean-difference vector at decode time to a demonstration-free context destroys the instructed format (0.000 vs. 0.425 under a norm-matched random vector). But no linear direction we tried *erases* it, even though two fitted projections drive the layer's linear code to chance (Appendix I). The code is low-rank, not redundant, and the behavioral decision does not consume the linearly decodable component. We can decode the mode and we can install it. Neither shows what carries it.

The carrier, localized in tokens. What direction-level search could not find, position-level search does—on the second family, where a counterfactual patching instrument exists (Appendix K): transplanting per-position hidden states between two transcripts identical except for their format instruction, and reading out the instructed formats' log-probability contrast (Table 4). Patching only the chat template's turn delimiters (the `im_end`/`im_start` glue, with no instruction text and no demonstrated answer among them) carries 62% of the full-context patch. Every *content* subset carries at most 12%. Layer-block patches place the state in the lower half of the stack, and the crossed cell (glue positions, layers 7–13 only) still carries 42%. The format state rides the transcript's *skeleton*, into tokens the chat template already supplies, which is consistent with a literature on delimiter and low-information tokens serving as computation slots (Appendix B). It is also where the OLMo attention row concentrates (§4.3, Fig. 3), and closing it disrupts the demonstrated shape. Unlike the mass absorbed by attention sinks (Xiao et al., 2024; Gu et al., 2025), which is content-independent normalization slack, the glue state is content-bearing and transportable: moving it moves the behavior. The channel is cross-family. On OLMo, whose full-transcript effect is an order of magnitude smaller and not inverted, the glue-only patch carries all of it ($+0.092$, n=85 after skips, vs. the full patch's $+0.056$, n=34, with the unrelated-fact control at -0.011 in both). A periodicity-matched content control remains open.

4.6 Why context rot gets reported: the signatures

If the entropy collapse behind the context-rot reports were a generic depth effect it should survive on heterogeneous dialogue. It does not. On 150 organic WildChat conversations (Zhao et al., 2024), within-conversation entropy is flat with depth (median late/early ratio 0.99). Partitioning a second, 400-conversation run by its *own* inter-turn homogeneity separates the driver: homogeneous tertiles collapse (0.897 late/early) while heterogeneous ones are flat (1.001, n=134 each). Because homogeneous conversations are *longer*, length works against the effect. Across OLMo-2's (Team OLMo, 2024) post-training chain the collapse is mostly a downward *level shift* in baseline entropy installed by alignment, not a steeper slope (Appendix J).

The residual hazard in the localized case is *calibration*, not competence. On the DDXPlus streams logging per-token confidence (n=154 pooled cases, two models, both question orders), answer confidence rises $0.85 \to 0.95$ with fill ($r = +0.72$) at flat accuracy—and equally on *wrong* answers ($0.86 \to 0.95$, $r = +0.69$). The model's confidence in its wrong answers therefore rises by $\sim$10 points over a session (Fig. 6c), most consequential exactly where a homogeneous high-stakes stream makes the model surest.

5 Conclusion

With accumulation, displacement, competition and precedent separated inside one harness, each mechanism leaves a distinct signature:

- **Accumulation** produces a monotone reallocation of attention and a collapse of predictive *entropy*, but no detected loss of accuracy and no loss of instruction adherence—the with-length decline that motivates the rot framing (Liu et al., 2023; Hong et al., 2025; Laban et al., 2025) does not surface in these streams.
- **Displacement** produces a large accuracy cost that is reproduced in full by removing the evidence's attention mass, graded in that mass, and positional in tokens. This is the one mechanism here that is attention dilution proper—the one for which attention-side steering and positional recalibration (Zhang et al., 2024; Hsieh et al., 2024a) act on the variable that actually mediates.
- **Competition** produces a further cost while the evidence's attention barely moves: interference that raw context length does not index (cf. Wang and Sun, 2025), carried instead by attention to the competing instances themselves, under 1% of the distribution.
- **Precedent** erodes not accuracy but *compliance*, in a contest the instruction loses with its attention intact. This is the benign, self-generated counterpart of demonstrations overriding priors and instructions (Wei et al., 2023; Anil et al., 2024; Camassa and Shiller, 2026), consistent with instructions and demonstrations occupying distinct representational channels (Davidson et al., 2025; Hendel et al., 2023; Todd et al., 2024).

The literature's central claim is therefore correct but misattributed and undercounted. In these streams, context rot is not a decay with length: it is three mechanisms with different signatures, of which length is merely the usual way of causing all three at once. A needle buried far away among plausible alternatives moves the displacement and competition levers at once, which is why length-indexed benchmarks cannot separate them (Hsieh et al., 2024b), and attention-share accounts of multi-turn decline (Dongre et al., 2026) describe only the mechanism that is dilution proper. The precedent state itself, unreachable as a direction, is reachable as a place: the transcript's skeleton electing itself as native instruction storage (Darcet et al., 2024; Sun et al., 2024; Mu et al., 2023). Displacement, its mass sufficiency, the accumulation null, and the precedent contest replicate on a second model family. Competition is the mechanism that does not: it is architecture-indexed, its attention signature inverting on Qwen (Appendix K).

6 Limitations

Per-condition n is bounded by the window, so every interval here depends on the paired analysis of §3. The accumulation result is a *net* null with asymmetric precision (2.9 points coherent, 5.5 random), so we claim no decline surfaces, not that accumulation carries no latent cost. The competition result is one significant step rather than a gradient, its closure rescue leaves $\sim 40\%$ of the gap unexplained, and it is the one mechanism that does not replicate across families. The precedent results use one instruction family at n=40 per cell, and the mode's carrier localization is still confounded with positional periodicity and has no generation-level corroboration under closure. Every mechanism experiment runs to a 4k window (§3), which is what makes the arms comparable and also what bounds the claims. We order the mechanisms at 4k and do not measure how their weights move at the 128k windows the phrase "context rot" usually names, though displacement is defined by attention share rather than token count and so carries a prediction at any length. The accuracy null holds for localized tasks, and length-dependent single-task settings are out of scope by design. Appendix A gives these in full.

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A Limitations in full

This expands the summary in §6.

Statistical. Per-condition n is bounded by the context window, so every interval here depends on the paired analysis of §3. Resampling the arms independently inflates each one about $2.5\times$, enough to report a true effect as null. The accumulation null is a *net* null with asymmetric precision (2.9 points coherent, 5.5 random), and the random-subject control removes subject-level predictability, not format or protocol learning—so we claim no decline surfaces, not that accumulation carries no latent cost, and the in-context-learning reading of that flatness is the consistent interpretation, not an isolated one.

Displacement. The necessity direction of the mediation remains only partially established, and the per-head pattern test narrows the explanation without closing it: head-uniformity is excluded, but the clamp still spreads each head's restored mass uniformly over the span's tokens, so a within-span token-pattern account and a non-attention positional channel are not separated.

Competition. The competition result is one significant step rather than a gradient: mild overlap is indistinguishable from none, and we do not claim cost rises with confusability. The near-duplicate arm also changes the selection regime, so overlap is implicated by the within-experiment fit and the closure rescue rather than the arm contrast alone. The competitor-closure rescue is substantial, not total: the point estimate leaves $\sim$40% of the gap unexplained (residual not distinguishable from zero). The measured-hot-set closure eliminates only the high-attention form of instrument slack. A low-mass non-verbatim channel (paraphrase without literal mention) is not excluded, and the hot-set ranking uses the all-layer mean row, so a ranking indexed to specific layers or heads could target differently. The instruction-canary null is a floor effect. The WildChat partition is observational, with summary values from teacher-forced analyses and no uncertainty on the tertile contrast.

Cross-family. The causal program ran in full on one model family. The Qwen2.5-7B replication (Appendix K) reproduces displacement, its mass sufficiency, the accumulation null, the clamp's compliance collapse, and the precedent ordering, but not the competition penalty, so that mechanism rests on one family—and the low-share compliance dissociation is OLMo-specific (Qwen's arm shares straddle its own causal threshold).

Precedent and the mode. The precedent results use one instruction family, graded by a rule-based checker at n=40 per cell. The filler streams co-vary in genre, structure, and reply length, so the applicability ordering rests on the stream contrast together with the one-example and reversal experiments. Window-split closure rules out attention to the exemplar answers as what sustains *or* installs the mode, but only for that span family: installation through attention to other context spans is untested. The mode's carrier localization has three open edges. The glue subset is confounded with positional periodicity (the random control is size-matched, not periodicity-matched). The patch readout is a teacher-forced log-probability contrast with no generation-level corroboration under closure (Appendix K). And the cross-family statement is about the *channel* only—OLMo's glue-only patch carries its full effect, but that effect is an order of magnitude smaller than Qwen's and the driver carries no per-cell interval, so no magnitude comparison is made. The mode also remains unlocalized as a *direction*: no linear direction we tried erases it.

Scope. The accuracy null holds for localized tasks. Genuinely length-dependent single-task settings are where dilution is the point, and are out of scope by design. Every mechanism experiment runs to a 4k window, which bounds the ordering to that regime (§6).

B Background and related work

Length as the suspected cause. Most evidence for context rot comes from setups where the answer-bearing information sits inside a single long input. Liu et al. (2023) showed that accuracy depends strongly on *where* in the context the relevant passage sits, dipping when it falls in the middle. Levy et al. (2024) held a reasoning task fixed and padded the input with task-irrelevant text, finding degradation well before the models' advertised context limits. Hong et al. (2025) ran a broad sweep over 18 models and found that reliability falls with input length even on tasks as simple as retrieval and replication, and that the fall is not uniform: it depends on how similar the needle is to the question, and on whether distractors are present. Hsieh et al. (2024b) show the benchmark-level version: near-perfect needle scores coexist with sharp degradation on harder long-context tasks, so claimed context sizes overstate usable ones. Laban et al. (2025) report a large multi-turn degradation, but locate its cause in the model committing early to a wrong reading and failing to recover—not in length as such.

Confusable context as a separate suspect. A second literature manipulates the *content* of the surrounding context rather than its length. Shi et al. (2023) insert an irrelevant sentence into grade-school math problems and observe a sharp accuracy drop. In retrieval-augmented generation, Cuconasu et al. (2024) find that documents retrieved as relevant but not containing the answer act as distractors and degrade accuracy. Wang and Sun (2025) show accuracy declining log-linearly as interfering similar updates accumulate, independent of raw context length—the nearest neighbor to our competition result, with no attention instrumentation. Our contribution is to locate the cost: it runs at held-constant *evidence* attention mass and is carried by attention to the competing instances themselves (§4.3).

Instructions versus demonstrations. Demonstrations can override a model's trained priors (Wei et al., 2023) and, at scale, its safety training (Anil et al., 2024). The same lever is framed as pure benefit in many-shot prompting (Agarwal et al., 2024). Hardcoded assistant turns exhibiting a competing pattern flip instruction-following into pattern-following (Camassa and Shiller, 2026), system-prompt adherence fails under conflicting or adversarial user turns (Mu et al., 2025), and instruction-derived and demonstration-derived task representations are measurably distinct (Davidson et al., 2025). Our eroding pressure differs from all of these: it is benign, self-generated, and contains no conflicting directive anywhere. It is precedent, not conflict or attack, and we identify the contest's causal weight and both of its reversals.

Attention as measurement and lever. Closest to our mechanistic question, Dongre et al. (2026) track attention from responses to the system prompt declining over 50 filler turns across four model families, show that hard-masking the instruction out of the window collapses compliance, and decode goal information from the residual stream at high AUC after attention has "closed". Three differences matter. Their goal-accessibility metric aggregates raw share, which falls by arithmetic as context grows (§2.2). Our token-normalized enrichment removes that decline entirely, and the corrected quantity is flat in eroding and non-eroding arms alike. Their causal arm is total ablation (attention necessary, which our system-span clamp confirms), while our clamp is graded and bidirectional, and the restoration direction (compliance recovered against 42 counter-exemplars) is an arm ablation cannot supply. And their filler is always inapplicable. Our applicability ladder shows the same model at the same near-zero share surviving or collapsing depending on what the filler demonstrates, a content dependence invisible to fixed-filler designs. On the steering side, Zhang et al. (2024) boost attention on profiled head subsets to improve compliance and Hsieh et al. (2024a) recalibrate a static positional attention bias globally. Our clamp is the measurement-grade counterpart (bit-identical no-op, solvable dose, span-targeted in accumulating multi-turn context), used to establish when attention mass is and is not the mediating variable. Position-driven attention structure is the content-independent backdrop to all of this: sink tokens absorb mass regardless of content (Xiao et al., 2024; Gu et al., 2025), and the causal mask plus RoPE produce distance decay by construction (Wu et al., 2025). Our clamps and token-normalized shares target content spans against that backdrop.

Compact task representations. In-context demonstrations compress into residual task vectors (Hendel et al., 2023) transported by identifiable heads (Todd et al., 2024). These are the no-attention-signature channel our precedent mechanism appears to use (§4.5), and our erase-null results bound what such compact readable structure can be assumed to do causally. The byte-exact format copying our pinned arms show (Appendix I) is the behavior induction heads implement at circuit level (Olsson et al., 2022). We implicate that circuit behaviorally and at span level, without head-level verification.

C Instrument validation

Attention capture. Shares are read from a reconstruction of each layer's attention at the final position (QK-norm where the architecture applies it, then RoPE, then the additive mask, then softmax), which avoids materializing $N \times N$ attention matrices. The reconstruction matches the framework's own `output_attentions` rows to max $|\Delta| = 1.1 \times 10^{-6}$ on OLMo-2 and 3.1×10^{-5} on a GQA architecture (Qwen-2), both in fp32. The capture applies the additive attention mask, which is what makes it valid *under* a mask-based intervention: a capture that ignores the mask still matches on plain forwards (a causal mask's last query row is all zeros) while silently reporting unclamped attention during a clamp.

Clamp. At scale 1.0 the clamp is bit-identical to the unhooked model (max $|\Delta|$ logits $= 0.0$), because it adds b=0 to the existing mask rather than reimplementing attention. The bias is uniform across heads, so a span's aggregate share is hit by bisection on b. All solved levels in this paper land within $\sim 10^{-4}$ of target. Setting a span's scale to 0 fills its key columns with the mask minimum—closure, the same operation as a causal mask. Interpretation is restricted to biases well clear of near-ablation: levels requiring -4.7 to -6.1 nats produce degenerate behavior (the model answers "A" on 103–110 of 110 items) and are excluded.

D Task and filler specifications

Probe construction. DDXPlus probes are 5-option differentials with the vignette as evidence. Options are shuffled per probe: DDXPlus lists the differential in rank order, so unshuffled gold is the letter "A" in 71.4% of probes, and arms that differ in letter bias then differ in accuracy for free. Items with fewer than 5 options are excluded (26.7% of raw cases, some answerable from a single option without the vignette). Generation budgets are per-protocol: the accuracy arms (§4.2–§4.3) generate 32 new tokens. The compliance arms (§4.4) generate 128 for the probe reply (fillers up to 200), because shorter caps truncate the reply component that formats put last, biasing exactly the metric under study. The probe and steering runs (Appendix I) generate 256.

Format instruction and grader. The clinical template requires `ANSWER:` with a letter and `SUPPORTING:` with at least two findings quoted from the vignette. Compliance requires the template. Accuracy is scored separately and more permissively (a letter offered in prose, or alone on the reply's first line, is scoreable for accuracy while remaining non-compliant). The grader was validated against real generations from every condition. Graders written against imagined output formats systematically mis-score exactly the conditions where the treatment changes the output format.

Filler streams. `code`: 30 synthetic request→Python-function tasks on one fixed template. `gsm8k`: grade-school math answered in the model's own worked prose. `mmlu`: 4-option MCQ (medical subjects excluded) answered with a bare letter. Filler replies are the model's own (up to 200 tokens, mmlu answers truncated to the letter), so each arm demonstrates the shape the model itself produces for that stream. The arms are compared at matched fill, not matched depth: mmlu accumulates ~ 0.018 fill per turn against code's ~ 0.056.

Canaries. The system-clamp arms use a three-part canary instruction (reply prefix marker, suffix tag, and a forbidden phrase). The adherence arm uses the prefix and suffix pair. The forbidden-phrase canary is satisfied trivially by short replies, so two canaries are informative.

E Example transcripts and generations

Both examples are the first probe of their committed runs, reconstructed byte-exactly from the run's seed and validated against the stored artifacts (token counts, pathology, gold). The generations are quoted verbatim from each run's stored `turns.csv`, truncated where the 32-token generation cap truncated them. Transcripts are abridged, and [...] marks elided text.

Displacement (§4.2). Session 0's first depth-21 probe: gold D (acute pulmonary edema), fill 0.50 in every arm. The transcript accumulates 21 answered MMLU filler exchanges. The vignette and the question are byte-identical across arms and only the vignette's position differs—in `local` it is the user turn directly before the question, in `back_10` the same turn sits ten filler exchanges earlier.

> `[user]` You are answering a mixed stream of questions. Some are multiple-choice knowledge questions; some describe a patient. Answer each one as it arrives.
>
> `[assistant]` Understood.
>
> `[user]` Which of the following ions CANNOT be used as a dopant in germanium to make an n-type semiconductor? A) As B) P C) Sb D) B. Reply with only the letter...
>
> `[assistant]` D
>
> [... 19 more filler exchanges ...]
>
> `[user]` Patient: 76-year-old Female. Chief complaint: Are your symptoms worse when lying down and alleviated while sitting up. Symptoms: pain (tedious, sensitive, heavy) in the lower and posterior chest, intensity 9, radiating to biceps, chin, jaw, shoulders; shortness of breath; bouts of choking that wake you up at night; swelling in ankles, calves, tibia [...] History: fluid in the lungs; dialysis; high blood pressure; heart attack or angina; heart failure [...]
>
> `[assistant]` Noted.
>
> `[user]` For the patient described earlier, most likely diagnosis: A) Pulmonary embolism B) Possible NSTEMI / STEMI C) Anaphylaxis D) Acute pulmonary edema E) Pericarditis. Answer:

In `back_10` the vignette turn and its `Noted.` sit above ten of the filler exchanges instead of none. The generations for this item:

> `local:` "D" (parsed, correct)
>
> `back_10:` "Given the symptoms described for the patient, such as sudden onset of severe chest pain that is worse when lying down and alleviated by sitting up, shortness of" (unparsed, wrong)

The displaced reply paraphrases the vignette accurately, so the model is not blind to the evidence, but it never commits to a letter. `back_2`, `back_5` and `back_20` fail the same way on this item. This is the unparsed failure mode whose rate rises with fill. The ladder survives restriction to parsed responses (§4.2), where the failures are wrong letters instead.

Competition (§4.3). Probe 0: gold E (viral pharyngitis). Every arm holds the vignette at the current query (distance 0) and accumulates eight answered DDXPlus cases. Only option overlap differs. A representative `near_dup` context case—four of its five options are the probe's, its gold is not:

> `[user]` Patient: 64-year-old Male [...] Symptoms: [...] diffuse muscle pain; loss of appetite; nasal congestion or a clear runny nose; a cough. History: smokes cigarettes; is immunosuppressed [...] Most likely diagnosis: A) Influenza B) Bronchitis C) Viral pharyngitis D) HIV (initial infection) E) URTI. Answer:
>
> `[assistant]` A
>
> [... 7 more such cases ...]
>
> `[user]` Patient: 13-year-old Male. Chief complaint: nasal congestion or a clear runny nose. Symptoms: burning pain in both tonsils, thyroid cartilage, pharynx, under the jaw, intensity 4; fever; nasal congestion; a cough. History: contact with a person with similar symptoms; lives with 4 or more people; is immunosuppressed [...]

`[assistant]` Noted.

`[user]` For the patient described earlier, most likely diagnosis: A) Bronchitis B) Acute laryngitis C) Influenza D) URTI E) Viral pharyngitis. Answer:

The generations for this item:

`disjoint:` "E" (correct)

`random:` "E" (correct)

`near_dup:` "D The patient's symptoms include a cough, fever, and nasal congestion, which are common in upper respiratory tract infections (URTIs). The fact that the" (parsed, wrong)

Under near-duplicate context the model picks URTI, an option that recurs in six of the eight context cases, over the gold, and justifies it from symptoms the context's cases share. With the same vignette over disjoint or random context it answers correctly.

F Statistical procedures

All intervals are case-resampled bootstrap 95% CIs at 10,000 draws (joint-fit and slope coefficient intervals at 4,000). Clamp and arm contrasts are *paired* over shared items. The paired analysis cuts interval half-widths $\sim 2.5\times$ against independent resampling on the same data. Joint fits are linear probability models (accuracy in percentage points is the quoted scale) with case-resampled coefficient intervals. Probe analyses use leave-one-out cross-validation at episode level with 200-shuffle permutation nulls. The overflow guard skips items that would not fit whole (prompt + generation headroom) and reports per-arm counts: $0/192$ in the distance sweep, 4 skipped $+15$ starved in competition (and the identical panel in the competitor-closure run), 0 in the mmlu erosion arm, and 7 in the code arm, all at its deepest depth, so that ladder is interpreted to depth 12 (fill 0.778).

G Per-layer and per-head structure

Attention share is a mean over a layer's 32 heads, and a mean can hold still while heads cancel, so we measured all $1{,}024$ heads under both accuracy-costing mechanisms.

Under displacement at layer 24, **every head loses evidence mass** from `local` to `back_20`, each significant at Bonferroni, so nothing is hidden by averaging. The loss is close to uniform in proportion: mean fractional drain 0.689 (sd 0.147), uncorrelated with how much mass a head held (r=0.08). A single uniform odds-scale reproduces 58% of the per-head pattern, which weakens (without eliminating) the reading of §4.2's partial necessity as pure instrument limitation.

Under competition at the same layer the net is 0.0003 while heads move 0.0026 against a paired sign-flip null of 0.0004 ($p \leq 0.0005$, 2,000 permutations), with 19 of 32 heads individually significant: a small reallocation, about 6% of the evidence's local share, rather than inaction. Finally, heads that concentrate on the evidence do exist (255 of $1{,}024$ attend it more than its token share warrants, up to 0.626 for one head on a span that is 0.102 of the context), but none of them are at layer 24, whose mean evidence share (0.0408) is among the lowest in the model (vs 0.183 at layer 3 and 0.143 at layer 16). Conclusions about head specialisation drawn from a single layer do not generalise, and we draw none.

H Additional displacement analyses

Table 3: **Distance is the variable that matters** (OLMo-2-Instruct, n=192 per arm, chance $= 0.200$, fill 0.688 in every arm), the numbers behind Fig. 2a. Evidence attention share is measured at layer 24 on the same forwards. Case-resampled bootstrap 95% CIs.

arm	`local`	`back_2`	`back_5`	`back_10`	`back_20`
accuracy	**0.464**	0.359	0.292	0.250	0.276
cost vs `local`	—	+0.104	+0.172	+0.214	+0.188
95% CI	—	$[+.005, +.203]$	$[+.073, +.266]$	$[+.120, +.307]$	$[+.094, +.281]$
evidence share @L24	0.0408	0.0363	0.0320	0.0195	0.0124

Tokens versus turns. A partial $2{\times}2$ in (gap turns) $\times$ (filler length), total context equalized by padding, dissociates them: two arms matched on *tokens* have effectively identical evidence share (0.0104 vs 0.0108) despite a $4\times$ difference in turn count, while at matched turns share falls $0.0288 \rightarrow 0.0104$ as the gap grows $446 \rightarrow 1684$ tokens—a 64% cut in the evidence's attention that costs at most 9.4 accuracy points ($+0.026$ $[-0.042, +0.094]$). Share regresses significantly on gap tokens and not on gap turns. Attention drain is a function of token distance, as RoPE decay predicts. Conversation structure is not what drains it.

The current query's floor. Accumulation drives the *current query*'s share from ≈ 0.35 down to ≈ 0.15, and it is natural to ask whether that stops short of a harmful level. It does not. Clamping the query span on cold-start contexts, accuracy is flat from the natural share (0.258, accuracy 0.545) down through 0.20 (0.509), then costs 16.4 points at 0.15 ($+0.164$ $[+0.036, +0.291]$)—exactly the share accumulation reaches. Accumulation stops *at* the edge of the flat region, not short of it. (Levels below 0.15 require -4.7 to -6.1 nats, near-ablation rather than points on a dose-response, and are excluded from interpretation.) The ladder re-denominates fractionally onto the all-32-layer mean share: natural 0.463, no cost above natural, flat at $0.775\times$ natural ($+0.045$ $[-0.018, +0.118]$), significant cost at $0.581\times$ ($+0.118$ $[+0.027, +0.209]$), the same structure as at layer 24. In those units accumulation lands at 0.202 over eight cases—*past* the first costly level, not at its edge. That accuracy nonetheless stays flat under accumulation, while the same share reached by clamping costs points, is itself evidence that in-context learning is actively holding accuracy up.

Figure 3: **Measured final-position attention, token by token** (OLMo-2, all-layer/head mean, log-scaled white→orange): windowed excerpts of the same probe's transcript with its evidence local vs. displaced ten turns back. Solid underline marks the evidence span, dashed the current question, and elided stretches are labeled with their token counts. The evidence's tint visibly fades when displaced (share 0.088 vs 0.052), and the hottest tokens in both arms are the chat template's turn delimiters—the concentration whose causal role §4.5 establishes. Built from stored capture rows, not an illustration.

I Precedent details

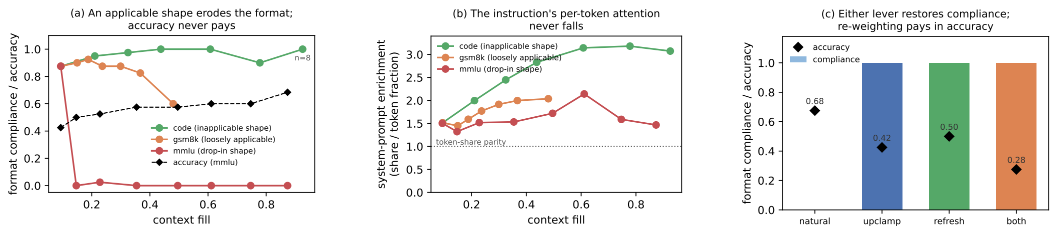

Figure 4: **Format erosion is a precedent contest, not attention starvation**, referenced from §4.4. (a) Compliance by fill for three filler streams whose demonstrated answer shapes differ only in applicability to the probe. mmlu accuracy (dashed) is unharmed through its total collapse. (b) The instruction's per-token enrichment is flat-to-rising in every arm, collapse included. (c) At depth 42, clamping the system span's share back up and restating the policy each restore compliance fully. Forced mass is paid for in accuracy (zero-sum), restatement is nearly free.

The system span's natural trajectory. Measured in the clamp experiment's own setup, the system span's share as prior cases accumulate is 0.166 (0 cases), 0.081 (1), 0.060 (2), 0.037 (4), 0.028 (6), 0.021 (8): an $8\times$ decline in eight turns from accumulation alone. Clamp ladders must be derived from the setup they are applied to. A share imported from a different context put two arms above natural, which were skipped rather than silently clamped upward.

One contrary example overrides the instruction entirely. With a single prior case whose answer exhibits the canaries, compliance is $3.00/3$ at every clamp level down to share 0.021. With a single bare-letter answer it is $1.00/3$ at every level including unclamped. 720 generations per arm, zero variance in reply length (8 characters in one arm, 1 in the other): the model reproduces the previous assistant turn's format exactly. Both arms are pinned, one at ceiling because the format is available from the transcript and one at floor because the contrary example already won, so the clamp has nothing to move. Only the neutral condition (no assistant turn anywhere) exposes the instruction's causal necessity reported in §4.4. A "no demonstration" arm whose prior turn answers with a bare letter is not an absence of demonstration. It is a counter-demonstration.

Exemplar closure. At depth 42, closing the demonstrated answer spans' key columns at generation time (scale 0, §2.3): all-answer closure 0.000 compliant, dose-matched partial closure 0.000, filler-question closure 0.132 (0.100 in the committed re-run, both consistent with pure renormalization of the surviving spans, the system's included), size-matched random-token closure 0.000. Accuracy moves within noise. Reading the exemplars at generation time is not what sustains the erosion. Figure 5 sets this against the competition rescue: the same intervention, opposite outcomes.

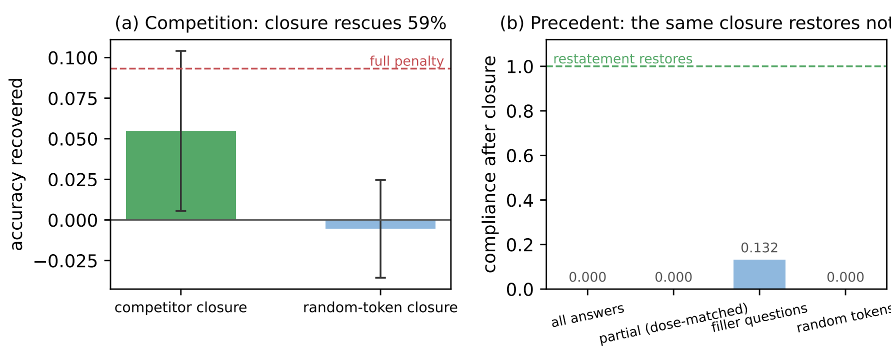

Figure 5: **One intervention, opposite outcomes.** The same scale-0 generation-time closure (a) recovers 56% of the competition penalty (paired $n{=}365$, bars are paired accuracy deltas with 95% CIs, dashed line marks the full in-run penalty), but (b) restores no compliance at depth 42 when applied to the demonstrated answer spans ($n{=}40$/arm, every closure arm sits at or near the eroded floor while restating the instruction restores 1.000). Competition is generation-time reading. The precedent mode is already in place by prefill.

Probes. Probe 1 (instruction presence): trained at depth 0 to separate the format system prompt from a token-length-matched neutral twin. Transfer AUC is 1.000 at every mmlu depth $0 \rightarrow 42$ (permutation nulls 0.49–0.50, $p<0.005$, $n=80/76$). The only depth trend is concentration: mean-over-layers AUC $0.985 \rightarrow 0.791$, peak layer drifting L1–2 $\rightarrow$ L11–14. Probe 2 (upcoming compliance): within gsm8k's mixed cells (same depth, fill, and filler, only the outcome differs, $n=80$), LOO-AUC 0.875 at stack layer 22 (null 0.485, $p<0.005$), crossing 0.8 from layer 18. Vector geometry: the mmlu and gsm8k mode vectors (deep minus shallow mean differences) have median cosine $+0.746$ across layers—two textually opposite demonstrated styles shift the residual stream in largely the same direction, with the caveat that generic deep-context components are common to both.

Steering and erasure. Install: the mean-difference vector at stack layer 21, applied at the final prefill position and every decode step at matched norm, destroys the instructed format in a demonstration-free context (0.000 vs 0.425 compliant under the norm-matched random control, natural 0.875), with replies reproducing the demonstrated registers verbatim. The collateral accuracy cost (0.175 vs 0.375) is consistent with deep-context components inside the mean-difference vector. A $3\times$ dose breaks generation for real and random alike—controls must be matched per dose. Erase: four strategies all null with clean controls—the mean-difference direction projected out at one layer and at eleven (the eleven-layer projection actively harms while its random control is harmless), and Probe 2's own reader direction projected out at its own layer, both in-distribution and transferred. Iterative re-probing on the captured states shows the layer-22 linear code is rank-$\approx$2 (AUC $0.875 \rightarrow 0.619$ after one fitted projection, 0.505 after two), so the erase nulls are not redundancy: the behavioral decision does not consume the linearly decodable component.

J Signature details

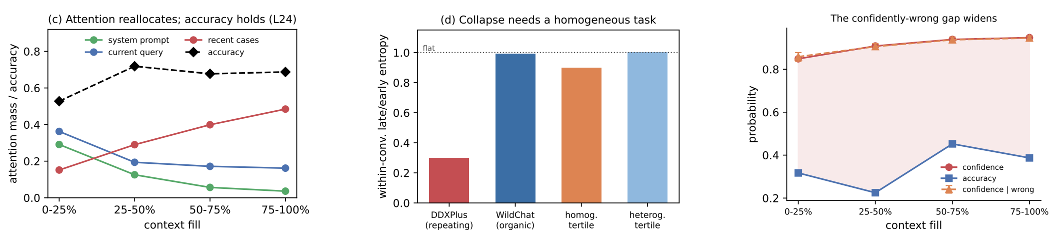

Figure 6: **The signatures track task homogeneity, and the residual hazard is calibration**, referenced from §4.6. (a) Attention reallocates monotonically off the system prompt and current query with fill while accuracy holds. (b) The entropy collapse appears only with a homogeneous repeating task, not on organic WildChat dialogue, and tracks homogeneity rather than length. (c) Confidence rises with fill on wrong answers as much as right, at flat accuracy.

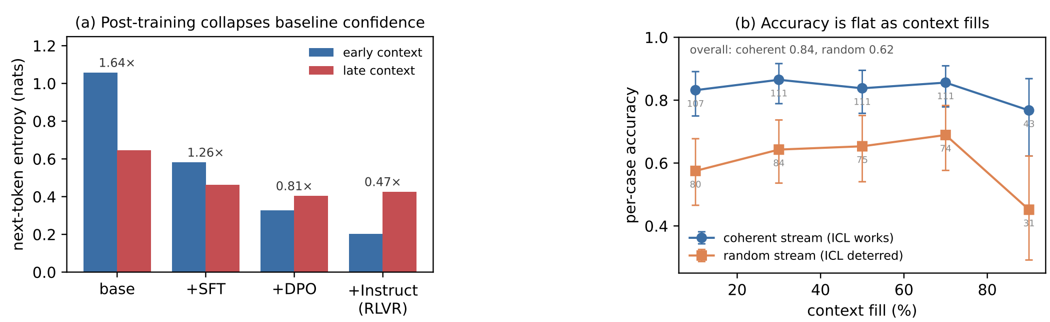

Figure 7: **Signature details.** (a) OLMo-2's post-training chain collapses baseline entropy stage by stage. The within-context ratio *falls* across stages, so alignment installs a level shift, not a steeper slope. (b) Accuracy is flat with fill in both the coherent and the random-subject stream. The random stream is lower throughout (harder subjects), not sloped. Error bars are Wilson 95% intervals with per-bin n printed. The final bins are the smallest (n=31/43) and their intervals overlap the preceding bins', so the endpoint movement is not resolved at this n.

Post-training chain. Replaying one probe across OLMo-2-7B's base→SFT→DPO→Instruct chain (identical cases and orders) shows early-context entropy collapsing monotonically, $1.06 \to 0.58 \to 0.33 \to 0.20$, with the largest single drop at base→SFT (Fig. 7a). The "collapses as context fills" framing is mostly a base-model ICL effect plus a floor: the within-context ratio *falls* across stages ($1.64 \to 0.47$), because aligned models start near an entropy floor.

WildChat. Median within-conversation $\mathrm{corr}(\text{entropy}, \text{depth})$ is -0.02 on organic dialogue (150 conversations, $2{,}039$ boundaries). The homogeneity partition is a separate 400-conversation run: tertiles of inter-turn TF-IDF similarity give late/early ratios 0.897 (homogeneous) vs 1.001 (heterogeneous), $n{=}134$ per tertile. Homogeneity correlates with the entropy slope (Pearson -0.141 at $p{=}0.005$, Spearman -0.103 at $p{=}0.039$, partial $r = -0.151$ controlling for length) and with conversation length ($+0.155$: homogeneous conversations are longer, so length works against the effect). The tertile medians are reported without intervals.

A proposed mechanism that fails causally. Clamping the boundary-detector SAE feature (Gemma Scope F90871, Lieberum et al., 2024) back to its clean level on held-out fatigued cases moves entropy the *wrong* way ($0.442 \to 0.465$ vs clean 0.293), leaves accuracy flat, and a random feature of similar magnitude perturbs entropy at least as much.

Attention and errors point the other way. Conditioning on fill, the cases OLMo-2 answers *wrong* have *higher* current-query attention than the ones it gets right (pooled $\Delta = +0.045$ $[-0.003, +0.097]$, $n{=}115$). The interval touches zero, but the sign is positive in every fill bin and at every probed layer, an observational and suggestive pattern rather than an established one. If it holds, errors co-occur with *fixating* on the query while correct answers spread attention onto the accumulated cases—the per-case fingerprint of the in-context learning that keeps accuracy up, with the "rot" and the within-context predictor of a correct answer pointing in opposite directions.

K Cross-family replication: Qwen2.5-7B-Instruct

The causal program was re-run on Qwen2.5-7B-Instruct (seed 42, 4k window, same drivers, panel constructions, and drop rules as the committed OLMo runs). The window matches the OLMo runs, so the cross-family contrasts below, the competition divergence included, are not window effects. Share readouts are all-layer pooled except the distance sweep's, which is layer-24-indexed like OLMo's. Displacement, its mass sufficiency, the dose-response, the accumulation null, the system-span clamp, and the precedent ordering reproduce. The competition penalty does not, and its attention signature inverts (Fig. 8). Details per experiment:

Table 4: **Counterfactual patching of the format-instruction state** (Qwen2.5-7B, `code` cell), referenced from §4.5. Full-transcript row $n{=}39$, bisection rows $n{=}23$. $\Delta\Delta$ = shift of the instructed-formats log-probability contrast, against a self-patch baseline.

patched positions	$\Delta\Delta$	share of full
full context, all layers	-2.488	100%
template glue (`im_end`/`im_start`)	-1.540	62%
assistant turns	-0.287	12%
user turns	$+0.165$	—
last 1/2/4 turns	$\leq \lvert 0.075 \rvert$	$\sim$0
size-matched random positions	$+0.016$	—
template glue × layers 7–13	-1.051	42%

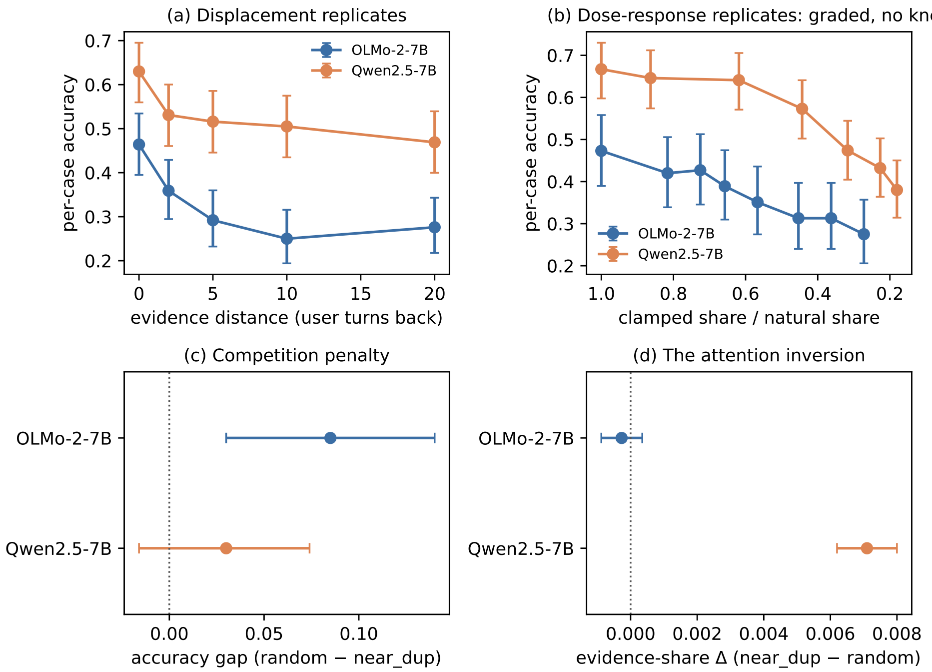

Figure 8: **The causal program replicates on Qwen2.5-7B, except competition.** (a) The distance ladder at matched fill (n=192/arm in both families). (b) The share→accuracy dose-response, x as a fraction of each family's natural share (OLMo layer-24-indexed with common subset n=131, Qwen all-layer with n=181–192): graded with no knee in both. (c) The competition contrast is significant on OLMo, not detected on Qwen. (d) The evidence's attention under near-duplicate context: flat on OLMo, *rising* on Qwen, with disjoint intervals—the family divergence. Whiskers in (a, b) are per-point Wilson 95% intervals. Panels (c, d) show the paired bootstrap CIs quoted in the text.

Displacement. The ladder reproduces fully monotone: $0.630 \rightarrow 0.531 \rightarrow 0.516 \rightarrow 0.505 \rightarrow 0.469$ (n=192 per arm, 0 overflow skips), every paired gap significant (`back_2` $+0.099$ $[+0.052, +0.146]$ through `back_20` $+0.161$ $[+0.094, +0.229]$). In the joint fit distance carries the coefficient (-0.0061 $[-0.0102, -0.0016]$) and fill is significant only with the *opposite* sign ($+0.330$)—byte-identical fill across arms cannot carry the ladder. Qwen answers with a bare letter (0.0% unparsed), so the ladder needs no parsed-only robustness arm: the truncation caveat attached to the OLMo ladder does not arise on this family.

Mass sufficiency and dose-response. Clamping `local` evidence to `back_20`'s share reproduces 107% of the distance gap (mass alone $+0.167$ $[+0.104, +0.229]$ against a gap of $+0.156$ $[+0.089, +0.224]$). The clamped arm is indistinguishable from `back_20` (residual -0.010 $[-0.057, +0.037]$). The share sweep is graded with no knee, still falling below the deepest natural share (0.667 at natural to 0.380 at share 0.0070), and its causal slope, $+10.4$ $[+7.6, +13.6]$ accuracy per unit share, matches the observational *correct∼share* fit ($+10.7$ $[+3.5, +17.6]$) from a separate design (the observational fit is on the distance sweep's layer-24 shares, the causal slope on all-layer shares, so the agreement is suggestive rather than unit-exact).

Accumulation and the system clamp. The random-subject stream is flat across its 4k window (fill slope -0.057 $[-0.238, +0.135]$, with the coherent stream at -0.090, also n.s.), and the adherence canaries do not decay. Clamping the system span's share collapses compliance at graded dose with a duty ordering, the reply's separately-graded obligations failing one at a time: reply-prefix canary first (at share 0.12), suffix next (0.09), the forbidden-phrase rule never (Fig. 9). Meanwhile accuracy *rises* under the clamp (-0.09 $[-0.16, -0.03]$ at the deepest level), the zero-sum reallocation that was directionally present but n.s. on OLMo.

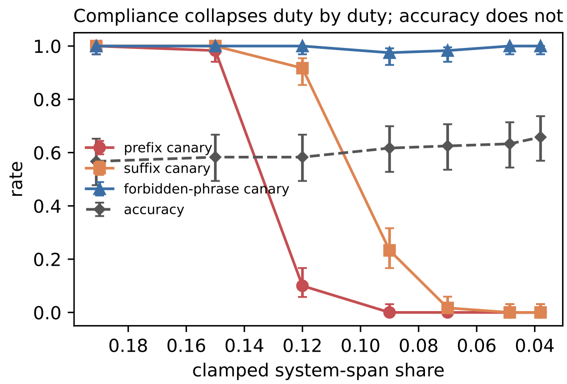

Figure 9: **Qwen's system clamp collapses compliance duty by duty.** Neutral-context clamp (n=120/level): the reply-prefix canary dies between shares 0.15 and 0.12, the suffix between 0.12 and 0.09, the forbidden-phrase rule never. Accuracy is flat-to-rising throughout. Whiskers are Wilson 95% intervals.

Precedent. The applicability ordering confirms: `mmlu` compliance $1.000 \to 0.000$ within three filler turns while `gsm8k` and `code` hold 1.000 through fill 0.84–0.92, accuracy flat throughout (overflow skips thin `code` to n=17 at its deepest fill). The demonstrated-answer reading signature reproduces (answer-over-question enrichment +0.87 to +1.49 in `mmlu` at every depth, negative in `code`—the erosion ordering). Two differences. Erosion is *staged*, not total: the `ANSWER:` marker dies immediately, the `SUPPORTING` duty survives to mid-depth—the same duty ordering the clamp produces causally. And restatement beats re-weighting: at depth 42, refreshing the instruction restores 1.000 while upclamping the system span restores only 0.733 (both at no accuracy cost). One dissociation does not carry over: Qwen's compliant arms sit at or above the share where its own clamp collapses the prefix duty and its eroding arm below it, so Qwen's erosion arms do not by themselves exclude a mass account the way OLMo's `code` arm (full compliance at half `mmlu`'s share) did.

Competition: the family divergence. On the same n=365 panel the penalty is not detected: `random` − `near_dup` = +0.030 [−0.016, +0.074], and the cross-family difference-in-differences is suggestive but not significant (+0.055 [−0.015, +0.125]). The attention signature, however, inverts with disjoint intervals: under near-duplicate context Qwen's evidence share *rises* (+0.0071 [+0.0062, +0.0080] vs. OLMo's +0.0003 null, orientation matched)—the model defends the evidence under confusability rather than merely not paying for it. Consistently, the same scale-0 competitor closure that recovers 56% of OLMo's penalty does nothing here (+0.019 [−0.025, +0.063], random-closure control 0.000). A masked-channel account is possible: by the dose-response slope the mass gain should be worth $\approx +0.07$ of accuracy, yet `near_dup` underperforms that prediction by ≈0.10—near OLMo's penalty. That account stacks coefficients across designs, so we flag it as interpretive. The closure null still constrains it: whatever the shortfall is, it is not carried by the competitor-mention channel that carried OLMo's cost. Qwen's low-competition arms also sit ≈0.10 above its own `local` (same-task context practices the probe's task where MMLU filler does not), so cross-family comparisons of arm *levels* inherit that practice benefit. Only within-family gaps are interpreted.

K.1 Counterfactual instruction patching and the carrier bisection

This is the instrument behind §4.5's token-level localization.

The instrument. Build two transcripts identical in filler and probe but differing in the system instruction, an `ANSWER:` prefix format against a one-line JSON format. Capture per-position hidden states from each. Then patch the recipient's prefill with the donor's states, and read out the two instructed formats' teacher-forced log-probability contrast. The deep `code` cell runs at depth 15 with the system span closed, in the A→B direction. The full-transcript patch is the stage-1 run at n=39, and the bisection cells are n=23 after one overflow skip.

The estimand, and why it is not the obvious one. $\Delta\Delta$ is a difference-in-differences against a *self-patch* baseline: the recipient's own open-captured states pushed through the identical procedure. It is not measured against the unpatched transcripts. The reason is that donor states are captured with the system span open, while a pure run's states form closed. Preflight showed that an unpatched baseline confounds that capture difference with donor identity, and the unrelated-fact control moved as much as the counterfactual patch. Against the self-patch baseline the same control is null.

Where the state lives. The full-transcript patch moves the contrast by $\Delta\Delta = -2.488$. Bisecting by position, every *content* subset is near null. The context's assistant turns carry -0.287, its user turns $+0.165$, and its last $1/2/4$ turns at most $|0.075|$. The chat template's turn delimiters alone carry $-\mathbf{1.540}$, 62% of the full effect, against a size-matched random-position control of $+0.016$. Those delimiters contain no instruction text and no demonstrated answer. Bisecting by depth instead, layer-block patches at all positions give -1.16 (layers 0–6), -1.59 (7–13), -0.45 (14–20), and -0.11 (21–27). The blocks overlap in what they carry, so their sum exceeds the full effect. Crossing the two, glue positions restricted to layers 7–13 still carry -1.051 (42%) while touching no content token and no layer above 13. Earlier waves that searched *inside* the turns, by role and by recency, had to come back null. The state rides between the turns rather than in them.

Specificity control. The control throughout is an unrelated-fact patch, the same instruction with one irrelevant clause changed, token-matched. It stays at $\leq |0.11|$ in every quoted cell except two, and both of those have counterfactual effects that are themselves near null: the last-2-turns cell at $+0.26$ and the random-position cell at -0.11.

Two caveats on the readout. Under closure Qwen emits no canonical format in free generation, in none of the 90 graded replies across both patch arms. The estimand therefore lives in the teacher-forced contrast alone, with no generation-level corroboration in this cell. Closure also drops the recipient's system-span identity out, which makes the two patch directions exact mirror images ($\Delta\Delta_{B\to A} = -\Delta\Delta_{A\to B}$), so the quoted direction is the one independent measurement. The full effect is also *sign-inverted* relative to the donor's instruction, in that A-donor states depress the A-format productions. The localization claim does not depend on that. The bisection asks *where* the state travels, not which way it pushes.